Final Project PS

PS1 : Digit Recognition

Develop an end-to-end handwritten digit recognition system using **Convolutional Neural Networks (CNNs)** that includes data preprocessing, CNN model design, training, validation, and performance evaluation. The model should be trained on a standard handwritten digit dataset and evaluated using appropriate accuracy and loss metrics to demonstrate the effectiveness of CNNs over traditional machine learning approaches. As the final test, write a single digit (0–9) clearly on a plain sheet of paper, capture its image using a mobile phone or webcam, preprocess the image to match the model's input format (grayscale conversion, resizing to pixels accordingly, and normalization), and pass it through the trained CNN to correctly recognize and display the digit.

Link of dataset : https://github.com/cvdfoundation/mnist

PS2 : Institute ID Card Dimension Detection

Capture an image of your **Institute ID card** using a mobile phone or webcam and determine its **dimensions in pixels** using basic image processing techniques. The task involves detecting the ID card from the image, identifying its boundary using **contours**, and calculating its width and height directly in pixel units without converting to real-world measurements.